

# WASP-2b

R-0181

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-2\_180901 · created 2026-07-16T19:18:12+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458362.6693 +/- 0.0023 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.13 +/- 0.02                   |
| Transit depth (Rp/Rs)^2                           | 1.68 +/- 0.53 [%]               |
| Orbital Inclination (inc)                         | 84.18 +/- 0.51                  |
| Airmass coefficient 1 (a1)                        | 1.06 +/- 0.011                  |
| Airmass coefficient 2 (a2)                        | -0.049 +/- 0.025                |
| Scatter in the residuals of the lightcurve fit is | 1.05 %                          |
| Best Comparison Star                              | #3 - [113, 74]                  |
| Optimal Aperture                                  | 4.02                            |
| Optimal Annulus                                   | 13.84                           |
| Transit Duration (day)                            | 0.0653 +/- 0.0069               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                        |
|---------------------------|----------------------------------------|
| Rp/R■ fit vs published    | 0.13 ± 0.02 vs 0.1326 (deviation 0.1σ) |
| Duration fit vs published | 0.0653 d vs 0.0737 d (deviation 0.92σ) |
| Mid-time O–C              | -12.8 min                              |
| Depth SNR                 | 4.9                                    |
| Residual scatter          | 1.05 %                                 |

### Light curve

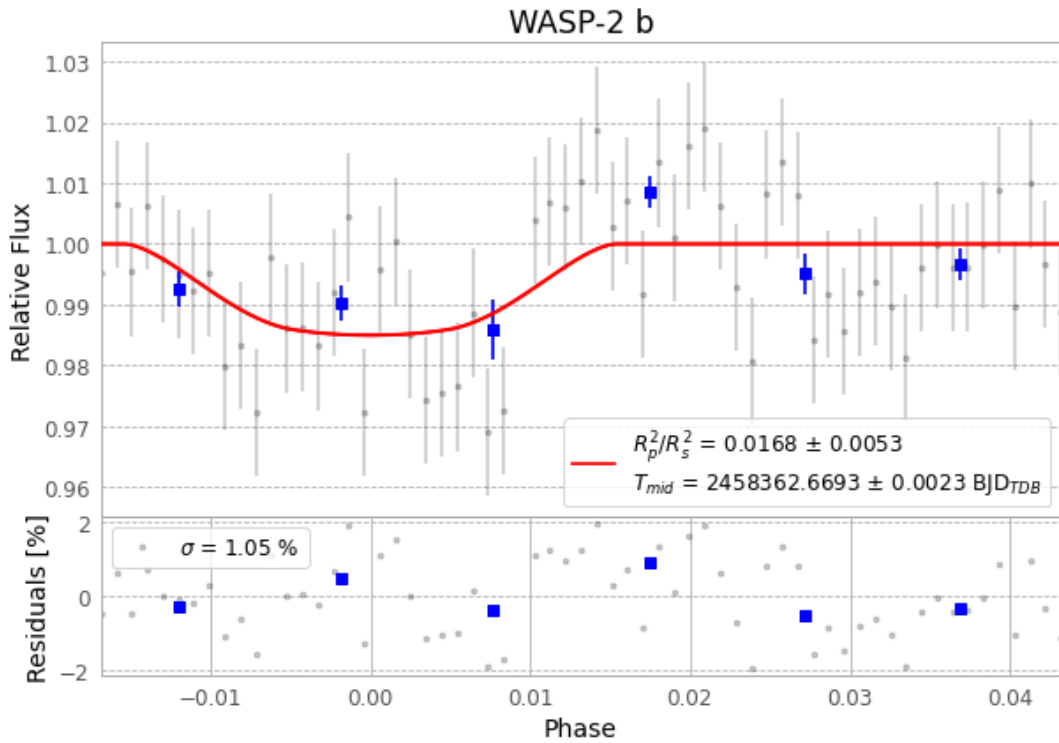


Figure 1 — FinalLightCurve\_WASP-2 b\_2018-08-31.png (SHA-256 25ad7cbfaf28cee7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 204], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 18b878f54810d20c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ee7e3ca

## Data provenance

63 FITS frames (41 MB), WASP-2180901030312.FITS → WASP-2180901060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180901.log (755437d4fafa...), FinalLightCurve\_WASP-2 b\_2018-08-31.png (25ad7cbfaf28...), FinalLightCurve\_WASP-2 b\_2018-08-31.csv (3174a64b2959...)

## Compute resources

Wall time 170.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:18Z.